El Hajj Malick Diagne

Robotics Engineering and Physics Student

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Objective

Aspiring robotics software engineer and researcher with an interest in control systems, embedded programming, and computational physics. Enjoys learning new skills using a project-based approach and understanding how technology works internally.

Education

Metro Early College High School — Columbus, OH
Aug 2023 – present
- GPA: 4.23 (weighted)
- Completed the DESIGN (engineering) pathway and a capstone project.

The Ohio State University, Early College Experience Student — Columbus, OH
Jan 2025 – present
- GPA: 4.00

Experience

The Metrobots (FRC Team 3324), Programming Manager — Columbus, OH
Aug 2024 – present
2 years 4 months
- Assisted with programming multiple robots which competed in First Robotics Competition (FRC) competitions.
- Led two simultaneous outreach projects to teach middle and high school students about digital electronics.
- Overhauled the team's website to use a modern web development framework (SvelteKit).

Metro Early College High School, High School Mathematics Tutor — Columbus, OH
Spring 2025 - 2026
- Tutored students in trigonometry, precalculus, and calculus work to assist students in grasping the course material.
- Answered questions for students when the teacher was unavailable, which included re-teaching concepts from previous courses.

Projects

Neutron-Star Matter Modeling Program — Spring 2026
- Developed an interactive web application using Python to model the structure of neutron-star matter under the mentorship of an Ohio State University physics professor.
- Utilized industry-standard data science and computational modeling libraries (NumPy, SciPy, Matplotlib, and Pandas).

Shift Register Control Panel — October 2026
- Designed and helped assemble a fully functional printed-circuit board (PCB) based around the 74HC595 shift register chip for a STEM outreach event.
- Coordinated with professional engineers to review and manufacture the PCB.

Skills

Programming Languages: C, Rust, Python, TypeScript, Kotlin, and Java

Platforms & Frameworks: Arduino, ESP32, Raspberry Pi, SvelteKit, and Android

Software: Onshape (3D modeling), KiCad (printed circuit board design), Linux (command-line interface), Git (version control), and Github